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New Class of Carbon-Nanotube Aerogel Electrodes for Electrochemical Power Sources**

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Carbon aerogels (CAGs) are novel ultralight materials with many interesting properties, such as low mass densities, continuous porosities, high surface areas, and high electrical conductivity. These properties are derived from the aerogel microstructure, which is a network of interconnected primary particles with characteristic diameters between 3 and 25 nm. Similar to carbon nanotubes (CNTs), the domain of applications of CAGs is enormous because of their unusual chemical and textural characteristics. For example, CAGs are promising materials for use as electrode materials for supercapacitors and rechargeable batteries, advanced catalyst supports for fuel cells, gas diffusion electrodes, adsorbents, chromatographic packing, thermal insulators, capacitive deionization, and molecular sieves, to name but a few. However, aerogels consist of agglomerate particles linked by covalent bridges (ladder structure). The contact between these particles and the space due to the pores unfortunately introduce a high internal resistance within the aerogel, which can be detrimental for electrochemical power source applications. In order to reduce the internal resistance, new strategies to introduce internal molecular nanocollectors (nanopathways for charges) such as CNTs absorbed in the porous matrix of the aerogel itself are being sought in our laboratory and elsewhere.^[1–5] It is also expected that the high mechanical resilience typical of CNTs would further strengthen the mechanical strength of the CAG in several applications.

Currently most aerogels and their composites are made by means of solution dispersion processes.^[4,6–11] We have recently reported the synthesis of carbonized polyacrylonitrile–multiwalled carbon nanotube (MWNT) composite materials processed from their dispersion in dimethylformamide (DMF) and explored their use as electrodes for electrochemical capacitors.^[4] The amount of MWNTs in the composite was varied from 3 to 12 wt %. The nanocomposites displayed high specific surface area (SSA) ranging between 670 and 710 m² g^{−1}. A capacitance as high as 218 F g^{−1} was obtained with an optimum composite containing 3 wt % MWNTs.

Larger amounts of the MWNTs diminished the effective portion of the active electrode material, resulting in the reduction of specific capacitance. Others have attempted reinforcement of CAGs with CNTs.^[1,3]

Dispersion preparation processes have proved rather difficult and still present considerable challenges, especially in the adhesion of nanotubes to the matrix material. Indeed, the effective utilization of CNTs in composite applications depends strongly on the ability to disperse the CNTs individually and uniformly throughout the host matrix without destroying their integrity or reducing their aspect ratio. The improvements that are required are better dispersion of CNTs within the aerogel matrix and better interfacial coating.

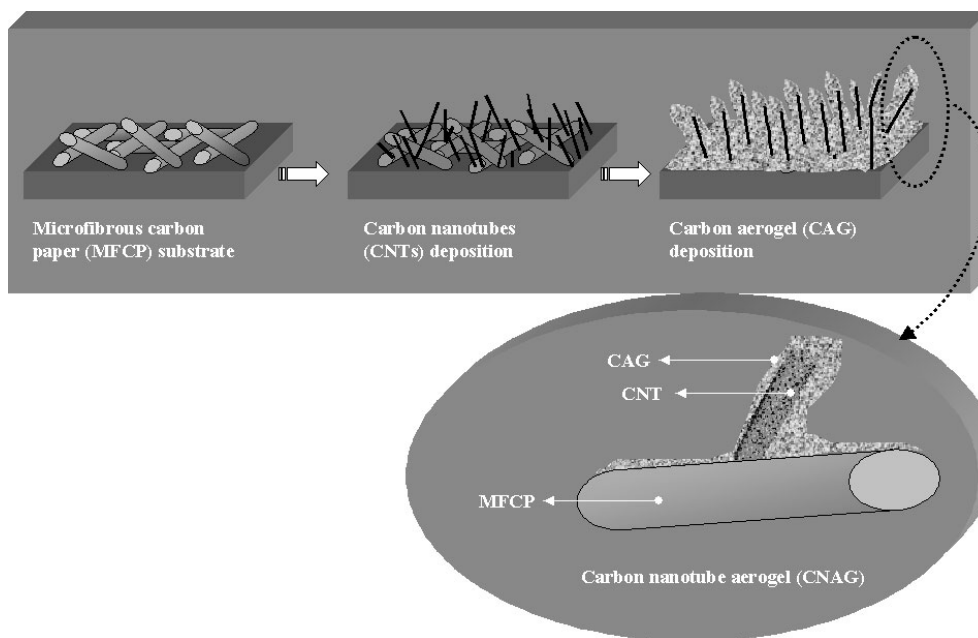
The new concept we introduce in this Communication is shown in Scheme 1. We propose the use of CNTs directly grown onto a microfibrinous carbon paper (MFCP) substrate. Afterwards, the CAG is synthesized on the CNTs. The flexibility afforded by this approach is the elimination of the need to control CNT concentration, thus allowing optimized CNT dispersion processes to be utilized for network reinforcement within the CAG. The CNT network is formed independently of the CAG, which makes the synthesis process very useful for making composites.

A further important feature of these nanocomposites is their lack of binder. Indeed, it is well known that practical electrodes are made of the active material, a conductivity enhancer, and a binder for the adhesion of the particles. The use of binders and/or other adhesives may occlude the surface of the nanotubes or electrically isolate some particles, whereas the conductivity enhancers may interfere with the response of the active material considered. Our carbon nanotube aerogel (CNAG) composites are fabricated directly on the MFCP substrate without any binder or conductive additive. Indeed, the MFCP is electrically conductive and can act as the current collector. Furthermore the MFCP is a synthetic, thin, cheap, lightweight matrix with pore spaces on the micrometer scale, and it is corrosion resistant, which makes the CNAG nanocomposite of even greater interest as binderless electrodes for electrochemical power sources, for instance.

Figure 1 shows scanning electron microscopy (SEM) and transmission electron microscopy (TEM) images of the pristine CAG. This material looks like a sponge with a highly connected structure (Fig. 1b). The TEM image reveals that CAGs consist of a network of particles arranged in an interconnected chain-like structure (Fig. 1c). Figure 2 shows SEM images of the prepared CNAG composites. Most of the substrate is covered with CAG (Fig. 2a–c) of texture similar to that ob-

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Scheme 1. Concept of making CNAG nanocomposites.

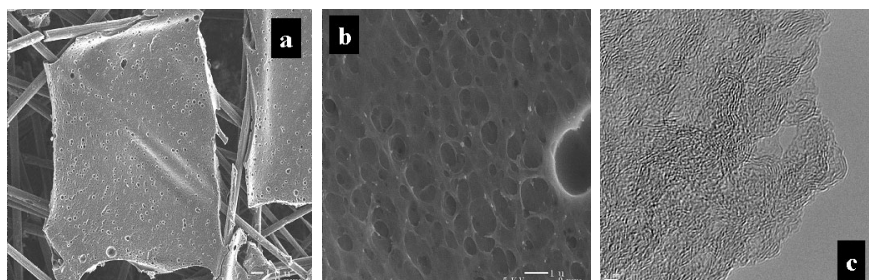


Figure 1. SEM (a,b) and TEM (c) images of CAG.

served for the pristine CAG shown in Figure 1. Beneath the CAG layer are seen the microfibers of the carbon paper substrate (Fig. 2d) uniformly coated with CNTs (Fig. 2e), which in their turn are nicely covered with CAG (Fig. 2f). This is clearly illustrated in the TEM photographs of Figure 3. It is of further interest to note the formation of spherical onion-shaped graphite encapsulating a nickel catalyst particle, as shown in Figure 3b. It has been claimed that metal particles such as Ni or Co can catalyze the formation of graphitic structures at relatively low temperatures.^[13–15] However, these onion-like graphitic structures have not been observed in our experimental conditions either for the pristine CAG (where Ni is absent) or for the pristine MWNTs (where Ni catalyst is present). The onion-like graphitic structures observed for the CNAGs are certainly an interesting point for further study. These studies will be the subject of a future report and should provide additional insight into this phenomenon. If Ni proved to be the catalyst for the formation of such onion-shaped graphite structures, it would definitely further expand the potential applications of these novel CNAGs to ferromagnetic applications. For example, carbon encapsulation of metal

nanoparticles can protect ferromagnetic particles from oxidation, making it an interesting magnetic medium. The CNAGs developed here could then constitute new structures for ferromagnetic elements such as Fe and Ni/Fe,^[16] or Co/Fe, which are already used for CNT synthesis.

Raman spectra of the pristine CAG (for comparison), the pristine MWNTs (for comparison), and the CNAG samples shown in Figures 2b,e are shown in

Figure 4. All the samples exhibit mainly two Raman bands, at around 1343 cm^{-1} (D band) and 1583 cm^{-1} (G band). The G band indicates original graphite features, whereas the D band is known to be related to disorder features of graphitic structures.^[17] It is well known that the R value, the relative intensity ratio of the D band to the G band, depends on both the degree of graphitization and the alignment of the graphitic planes of carbon materials.^[18] Also, the R value is sensitive to the ratio of the concentration of graphite edge planes and/or crystal boundaries to standard graphite planes, i.e., the lower the R value, the greater the amount of sp^2 (graphite) clusters on the sample. The intensity ratios of the D band to the G band (I_D/I_G) were determined and were found to be 0.58, 0.999, 0.925, and 0.900 for CAG, MWNT, CNAG (of Fig. 2e), and CNAG (of Fig. 2b), respectively. These results indicate that the new CNAG specimens are disordered carbon.

Atomic concentrations obtained by X-ray photoelectron spectroscopy (XPS) survey revealed that the relative ratios of the surface concentration of carbon to oxygen are 19.36, 9.5, and 8.17 for MWNT, CAG, and CNAG samples, respectively (Table 1). Table 1 also reveals that the amount of nitrogen is

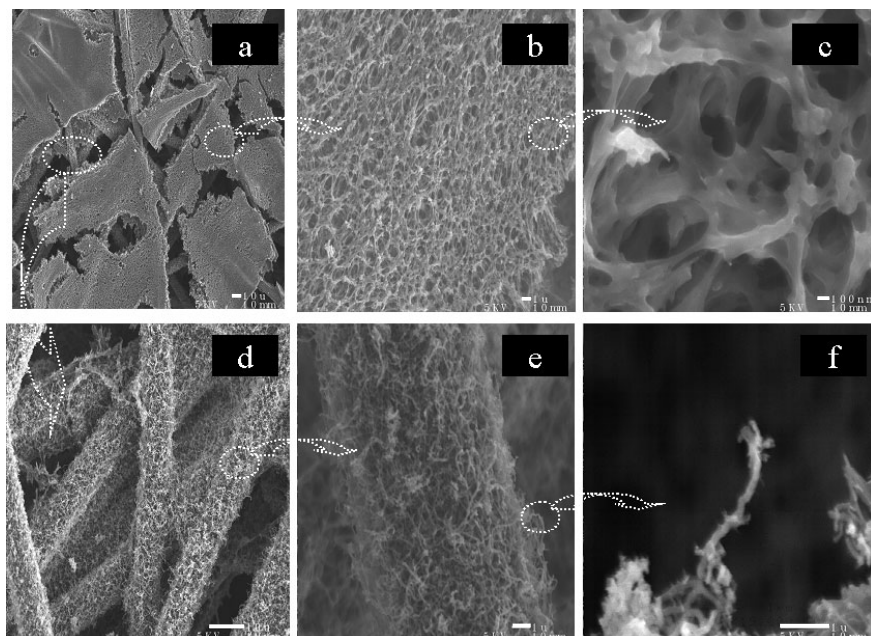


Figure 2. SEM images of CNAG nanocomposites.

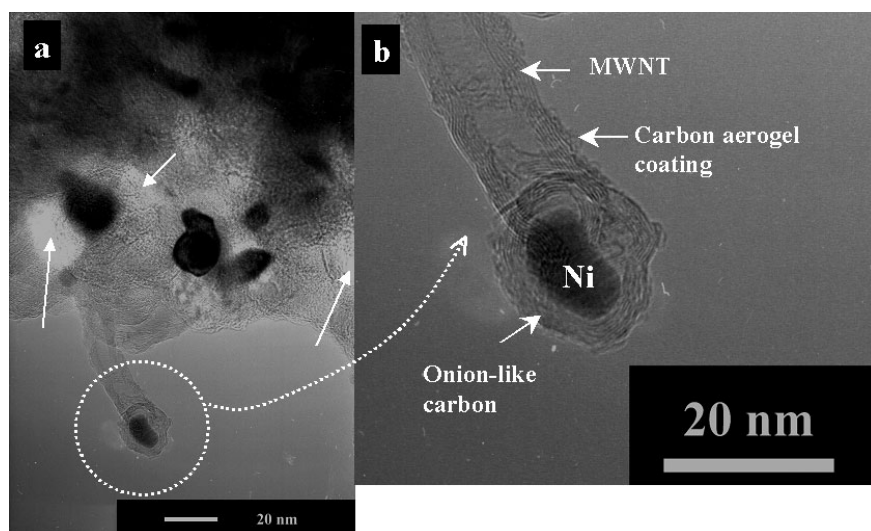


Figure 3. TEM images of CNAG nanocomposites. White arrows in (a) indicate pores.

lower in CNAG samples than in pristine CAG. XPS spectra for the CNAG sample are presented in Figure 5. The high-resolution C 1s core level peaks could be fairly deconvoluted into three major components centered around 284.4, 285.6, and 286.8 eV, which are attributed to C–C, and C–OH, and C=O functional groups, respectively, in accordance with literature data.^[19,20]

Measurements of the SSA and pore dimensions analysis are summarized in Table 2. The results reveal that the CNAG sample has a remarkably high SSA of $1059 \text{ m}^2 \text{ g}^{-1}$ and large pore volume of ca. $2.4 \text{ cm}^3 \text{ g}^{-1}$, which is more than six times that of the pristine CAG. Furthermore, it is seen that the com-

bination of MWNTs (mesoporous) and CAG (mostly microporous) gives the CNAG a textural porosity of almost of 50/50 mesoporous and microporous, which should be interesting for electrochemical capacitor applications. In fact, the micropores will participate in the charge storage processes, and wide pathways, that is, mesopores, will permit a fast accessibility of ions, thus a high energy at a high rate.^[21]

Preliminary testing to evaluate the potential of these newly developed CNAG composites for supercapacitors was carried out in 5M KOH electrolyte. Figure 6 shows typical cyclic voltammograms (CVs) for a CNAG electrode recorded at 20 mV s^{-1} scan rate. The CNAG electrode exhibited excellent capacitor behavior: the curve presents the typical rectangular shape expected for an ideal double-layer capacitor. Experiments at scan rates ranging from 5 to 100 mV s^{-1} were performed and results showed that the current increases with increasing scan rate with no significant degeneration of the shape of the cyclic voltammograms at the high scan rates. It is known that specific capacitance can be estimated well from constant current (CC) charge–discharge measurements.^[22] Thus, by means of CC charge–discharge experiments, it was found that CNAG delivered an outstanding high specific capacitance of 524 F g^{-1} , almost four times that of CAG (134 F g^{-1}). To the best of our knowledge this is by far the highest capacitance for carbonaceous materials ever reported in the open literature. For comparison, the highest capacitance obtained with activated carbon was 320 F g^{-1} .^[23]

In summary, we have developed a new class of CNAGs that differ significantly from those previously reported in the literature. The CNAG clearly demonstrated their great potential as electrodes for supercapacitor applications. The remarkable supercapacitive performance of the CNAG is attributed to the high

Table 1. XPS elemental analysis of properties of CNAG as compared with pristine MWNTs and pristine CAG.

Sample	C [at %]	O [at %]	N [at %]	C/N	C/O	O/N
MWNT	95.09	4.91	–	–	19.36	–
CAG	87.804	9.233	2.963	29.63	9.5	3.11
CNAG	87.3	10.68	2.28	38.3	8.17	4.68

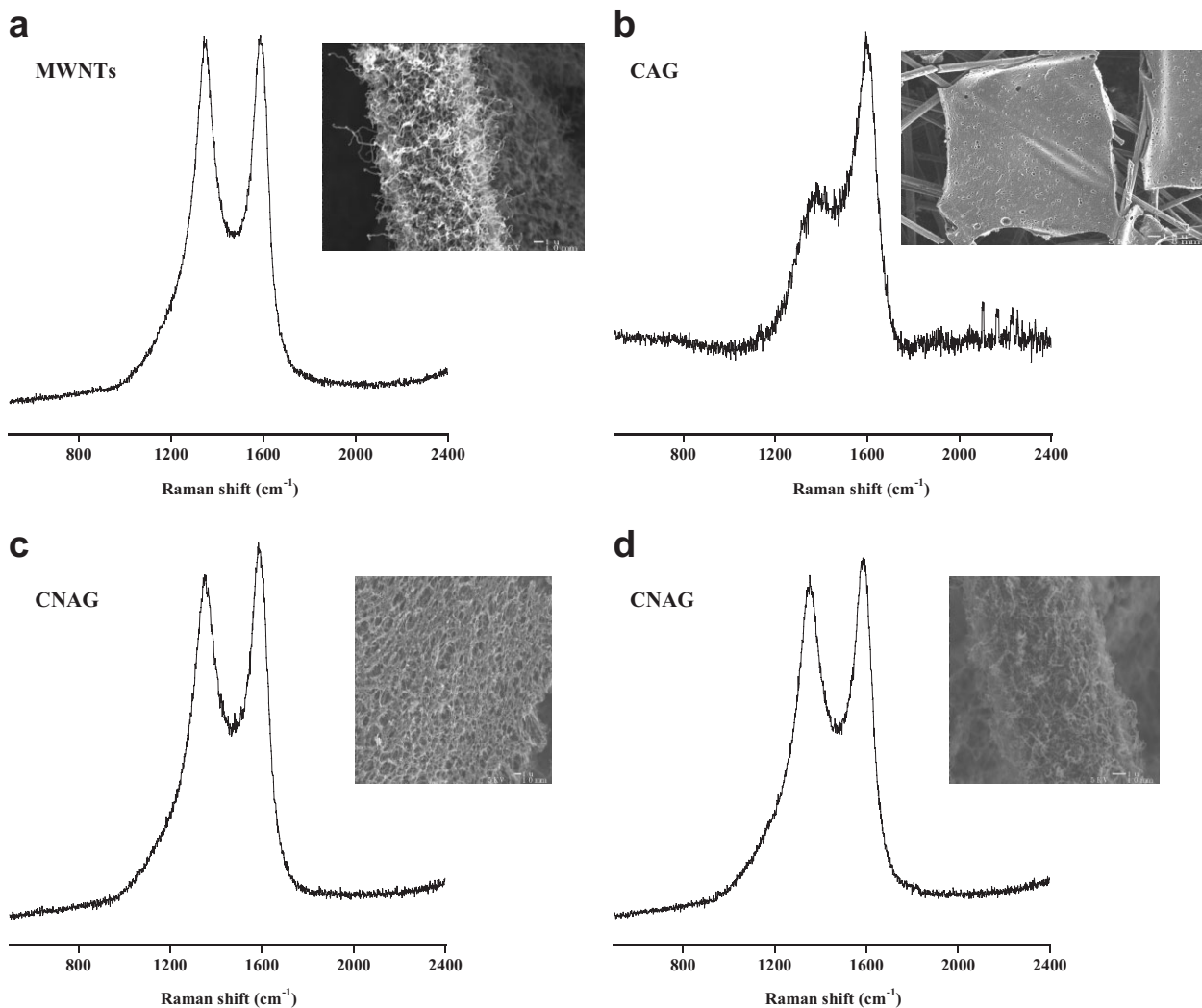


Figure 4. Raman spectra for as-synthesized MFCP-MWNTs (a), CAG (b), and CNAG (c,d). The insets are SEM images showing areas where micro-Raman measurements were taken.

Table 2. Textural properties [a] of CNAG as compared with pristine MWNTs and pristine CAG.

Sample	S_{BET} [m ² g ⁻¹]	S_{mes} [%]	S_{mic} [%]	V_{mic} [cm ³ g ⁻¹]	V_{mes} [cm ³ g ⁻¹]	P_V [cm ³ g ⁻¹]	P_D [nm]
MWNT	92.5	100	0	0	0.1951	0.1951	2.53
CAG	772	12.4	87.6	0.234	0.130	0.367	1.91
CNAG	1059	49	51	0.369	2.001	2.3697	1.41

[a] Brunauer–Emmett–Teller (BET) specific surface area S_{BET} , density functional theory (DFT) micropore surface area (S_{mic}) and volume (V_{mic}), mesopore surface area (S_{mes}) and volume (V_{mes}), pore volume (P_V), and pore diameter (P_D).

specific surface area, the special textural porosities, and the binderless character of the electrodes.

Furthermore, given these interesting physicochemical properties, these nanostructured CNAG electrodes present a distinct opportunity for creating novel nanocomposite materials for a wide range of electrochemical power source applications, such as insertion electrodes for lithium ion batteries and novel

electrode structures for fuel cells, either as catalyst-dopants or catalyst-surface supports.

Experimental

The synthesis of MWNTs on MFCP substrate was performed by chemical vapor deposition (CVD) and is reported elsewhere [12] The CNAG was prepared as follows: First polyacrylonitrile (PAN) powder was dissolved in a mixture of DMF and distilled water (84/16 mL) and the whole was heated until it boiled, at which point the aquagel was formed. Afterward, the MFCP/MWNT composite was dipped in the aquagel to allow this to deposit by coating. The resulting composite MFCP/MWNT/aquagel film was soaked overnight in acetone prior to the pretreatment and carbonization steps. Next, the resulting MFCP/MWNT/aquagel was pretreated in air at 215 °C for 20 h. This important step led to cyclization reactions and to a ladder structure. Subsequently, the sample was carbonized at 850 °C for 8 h in an inert atmosphere (argon), leading to the CNAG. Then, the CNAG samples are activated under carbon dioxide treatment using the following soak cycle. First the sample was heated from ambient temperature to 800 °C (over 1 h) then held at 800 °C under argon flow for 45 min. The atmo-

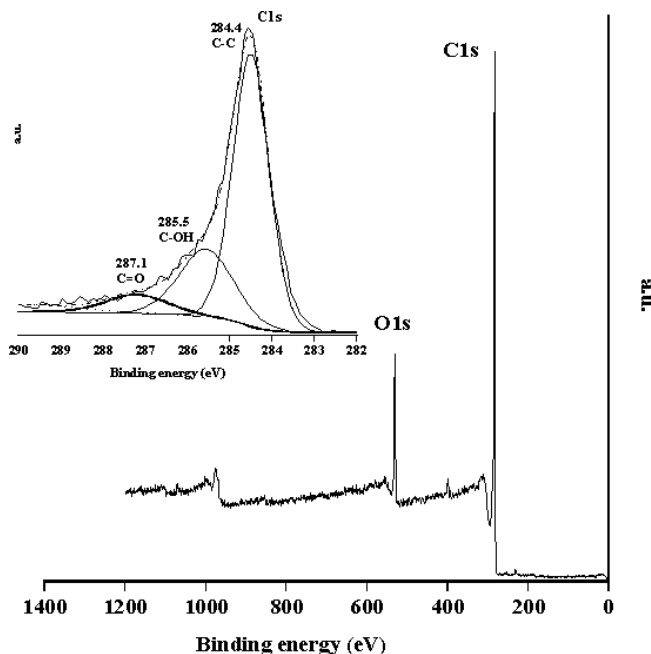


Figure 5. XPS high-resolution spectra (C 1s region) for CNAG nanocomposites.

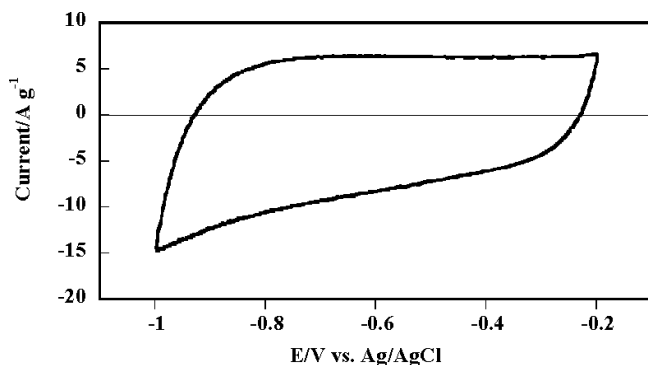


Figure 6. Cyclic voltammetry of CNAG in a 5M KOH electrolyte. Potential scan rate 20 mV s⁻¹.

sphere was then changed to a mixture of argon and carbon dioxide for 15 min and then maintained at 800 °C for 1 h under carbon dioxide. Finally, the atmosphere was switched to argon gas and the temperature was reduced to room temperature.

XPS was conducted for elemental composition of the material's surface and its chemical environment. The XPS study was performed with a VG Instruments Escalab 220i-XL surface microanalysis system equipped with hemispherical analyzer and Al K α X-ray source (1486.6 eV). Survey scans in the range 0–1000 eV were recorded at 100 eV pass energy with a step size of 1 eV. Core level spectra were obtained for C 1s and recorded at 20 eV pass energy with a step size of 0.1 eV. Curve fitting of the XPS data was carried out with casaXPS version 2.2.107. A semiquantitative evaluation of relative atomic surface concentrations was obtained by considering their corresponding sensitivity factors: C 1s (1.0), N (1.8), and O 1s (2.93). The binding energies were corrected for surface charging by referencing them to the designated hydrocarbon C 1s binding energy of 285.5 eV.

Micro-Raman measurements were taken with an RM 1000 Renishaw Raman microscope system equipped with a laser at 514 nm, a

Leica microscope, and an electrically refrigerated charge-coupled device (CCD) camera. The spectra shown here were obtained with 20 $\times$ magnification objective lenses. The laser output was 50 mW. Acquisition time was 5 min. The software employed for data acquisition and analysis was WIRE for Windows and Galactic Industries GRAMS/32TM. Ten scans were recorded to improve the signal-to-noise ratio. Spectra at three different points were taken for each sample to minimize problems arising because of the possible lack of sample uniformity. The Raman spectra presented are the representative measurements of the three points.

The surface morphology of the samples was examined by means of a scanning electron microscope (JEOL, JSM 6300F apparatus) operated at an accelerating voltage of 5 kV and a transmission electron microscopy (JEOL JEM-2100F) operating at 200 kV.

SSA was determined from nitrogen adsorption isotherms at 77 K (Autosorb-1 Quantachrome instrument) using standard Brunauer–Emmett–Teller (BET) theory. The micropore size distribution, surface area (S_{mic}), and volume (V_{mic}), and the mesopore surface area (S_{mes}), volume (V_{mes}), and size distribution were analyzed by density functional theory (DFT), provided by the Autosorb-1 for Windows 1.19 software (Quantachrome Corp.).

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